

**Real-Time Off-Tracking (Inner-Wheel
Difference) Early Warning for Articulated
Trucks:
From MATLAB Closed-Loop Simulation to
Embedded ESP32 Implementation**

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Abstract

Large articulated trucks exhibit significant low-speed off-tracking during turning maneuvers, where the trailer rear axle path deviates inward relative to the tractor path. This phenomenon is a major contributor to right-turn collisions with pedestrians and roadside objects. This thesis presents a complete research and engineering pipeline: (i) a single-unit vehicle closed-loop yaw control simulation with a geometric instantaneous-center-based off-tracking measure, (ii) an articulated tractor-trailer kinematic model with dual-loop PID steering control, (iii) a swept-area (danger-zone) model built from the trailer right-side body boundary, and (iv) a real-time embedded implementation on an ESP32-S3 platform that predicts the 1-second future swept area and checks millimeter-wave radar detections for intersection with the predicted danger zone. The system employs a receding-horizon prediction strategy in the tractor rear-axle body frame, enabling robust real-time decisions without global positioning or map-level localization. Experimental and simulation results demonstrate that the proposed geometry-based warning logic is computationally efficient and suitable for embedded deployment.

Keywords: articulated truck; off-tracking; swept area; kinematic prediction; PID control; mmWave radar; ESP32.

Acknowledgments

(Optional. Write acknowledgments here.)

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Nomenclature

Symbol	Description
A	Front axle (or front wheel reference point) of the tractor
B	Rear axle center point of the tractor (body-frame origin)
H	Hitch / kingpin point connecting tractor and trailer
T	Trailer rear axle center point
θ	Tractor yaw angle (heading)
θ_t	Trailer yaw angle (heading)
ϕ	Articulation angle between tractor and trailer
α	Tractor front steering angle
v_A	Tractor front reference speed
v_B	Tractor rear axle speed (kinematic projection)
l	Tractor wheelbase length (front axle to rear axle)
l_h	Tractor rear overhang (rear axle to hitch)
L	Trailer axle length (hitch to trailer rear axle)
w	Vehicle track width
ICC	Instantaneous center of curvature (single vehicle)
ICCT	Instantaneous center of curvature of trailer rear axle T
TF	Off-tracking distance defined by the line intersection method
Δt	Discretization time step
N	Number of time steps

Chapter 1

Introduction

1.1 Background

Low-speed turning of heavy trucks often produces a characteristic inward deviation of the trailer path relative to the tractor path, commonly termed *off-tracking* or *inner-wheel difference*. In urban right-turn scenarios, pedestrians or roadside objects may be outside the tractor’s immediate clearance region yet still be swept by the trailer body. This motivates a real-time early warning system capable of predicting the future swept area and checking whether obstacles fall into that area.

1.2 Problem Statement

Given real-time measurements of (i) steering angle α , (ii) articulation angle ϕ , (iii) vehicle speed v , and (iv) millimeter-wave radar detections on the right side, design an embedded algorithm that:

- (1) predicts a short-horizon swept area of the right-side truck body boundary, and
- (2) determines whether any detected obstacle lies within the predicted danger zone,
- (3) triggers a warning with robust temporal debouncing.

1.3 Contributions

This work contributes:

- A staged modeling pipeline from single-unit kinematic yaw control to articulated truck modeling.
- A geometric off-tracking metric based on instantaneous curvature centers (single-unit) and a line-intersection TF definition (articulated case).

- A time-discretized swept-area model using right-side body boundary points (hitch side and trailer rear axle side).
- An embedded architecture on ESP32-S3 using a receding-horizon prediction in the tractor rear-axle body frame, enabling map-free real-time warnings.

1.4 Thesis Organization

Chapter 2 presents the single-unit vehicle model and PID yaw control, and defines the instantaneous-center-based off-tracking measure. Chapter 3 extends the model to articulated trucks and introduces the TF off-tracking definition. Chapter 4 defines the swept-area (danger zone) generation. Chapter 5 details the embedded implementation, radar coordinate transformation, point-in-swept-area tests, and software architecture. Chapter 6 concludes and discusses future work.

Chapter 2

Single-Unit Vehicle Modeling and Closed-Loop Control

2.1 Kinematic Model

A low-speed kinematic (bicycle) model is used. Let A be the front reference point and B be the rear axle center. The heading is computed by the axle line:

$$\theta = \text{atan2}(y_A - y_B, x_A - x_B). \quad (2.1)$$

Assuming front reference speed v_A and steering angle α , the rear axle speed is approximated by:

$$v_B = v_A \cos \alpha. \quad (2.2)$$

Then the rear axle motion is:

$$\dot{x}_B = v_B \cos \theta, \quad \dot{y}_B = v_B \sin \theta. \quad (2.3)$$

The yaw-rate relationship is:

$$\dot{\theta} = \frac{v_A \sin \alpha}{l}. \quad (2.4)$$

2.2 Reference Heading Profile

A piecewise heading reference $\theta_{\text{ref}}(t)$ is designed for a full maneuver from 90° to 0° :

$$\theta_{\text{ref}}(t) = \begin{cases} \frac{\pi}{2}, & t \leq t_1, \\ \frac{\pi}{2} \left(1 - \frac{t-t_1}{t_2-t_1}\right), & t_1 < t \leq t_2, \\ 0, & t > t_2. \end{cases} \quad (2.5)$$

2.3 PID Steering Control

Define the heading error:

$$e(t) = \theta_{\text{ref}}(t) - \theta(t). \quad (2.6)$$

A PID controller generates the steering command:

$$\alpha(t) = K_p e(t) + K_i \int_0^t e(\tau) d\tau + K_d \frac{de(t)}{dt}. \quad (2.7)$$

To prevent wind-up and maintain physical feasibility, both the integral term and the steering output are saturated:

$$\int e dt \in [-I_{\max}, I_{\max}], \quad \alpha \in [-\alpha_{\max}, \alpha_{\max}]. \quad (2.8)$$

2.4 Wheelbase Constraint Correction

Numerical integration may introduce drift in the axle distance constraint $\|A - B\| = l$. A geometric correction step is applied to maintain:

$$\sqrt{(x_A - x_B)^2 + (y_A - y_B)^2} = l. \quad (2.9)$$

This correction improves numerical stability and ensures consistent kinematic geometry across the simulation horizon.

2.5 Instantaneous Center of Curvature and Off-Tracking (Single-Unit)

An instantaneous center of curvature (ICC) can be computed from the velocity normals. For each axle point $P \in \{A, B\}$ with velocity $\mathbf{v}_P = [v_{Px}, v_{Py}]^\top$, a normal direction is:

$$\mathbf{n}_P = [-v_{Py}, v_{Px}]^\top. \quad (2.10)$$

The ICC point O is the intersection of the normal lines:

$$A + s \mathbf{n}_A = B + t \mathbf{n}_B. \quad (2.11)$$

If the normals are not parallel, solve for s, t and obtain O . Radii are:

$$R_A = \|A - O\|, \quad R_B = \|B - O\|. \quad (2.12)$$

The single-unit off-tracking metric is defined as:

$$d_{\text{single}} = R_A - R_B. \quad (2.13)$$

2.6 Summary

This chapter established a baseline closed-loop simulation and a geometric off-tracking measure. The next chapter extends the modeling to articulated trucks, where off-tracking is amplified and requires a more expressive metric and swept-area reasoning.

Chapter 3

Articulated Truck Kinematics and TF Off-Tracking Definition

3.1 Articulated Geometry

The articulated configuration introduces a hitch point H and trailer rear axle T . Given tractor rear axle B and tractor heading θ , the hitch point is:

$$x_H = x_B + l_h \cos \theta, \quad y_H = y_B + l_h \sin \theta. \quad (3.1)$$

Let ϕ be the articulation angle and define trailer heading:

$$\theta_t = \theta - \phi. \quad (3.2)$$

Trailer rear axle point:

$$x_T = x_H - L \cos \theta_t, \quad y_T = y_H - L \sin \theta_t. \quad (3.3)$$

3.2 Dual-Loop PID Concept

To generate stable turning trajectories in simulation, a dual-loop concept is applied:

- Outer loop: heading tracking for tractor $\theta \rightarrow \theta_{\text{ref}}$.
- Inner loop: articulation suppression/regulation $\phi \rightarrow \phi_{\text{ref}}$ (typically $\phi_{\text{ref}} = 0$).

The steering command can be expressed as:

$$\alpha = \alpha_\theta + \Delta\alpha_\phi, \quad (3.4)$$

where each term is computed by a PID controller with saturation to enforce actuator and safety limits.

3.3 TF Off-Tracking Definition via Trailer ICC and Line Intersection

Unlike the single-unit radius difference, articulated off-tracking is defined using a geometric intersection method (TF metric):

3.3.1 Trailer Rear-Axle ICC (ICCT)

Estimate trailer rear-axle speed $\mathbf{v}_T$ and trailer yaw rate $\omega_t = \dot{\theta}_t$ using finite differences and smoothing/low-pass filtering. The signed curvature radius is:

$$R_t = \frac{\|\mathbf{v}_T\|}{\omega_t}. \quad (3.5)$$

Define the left normal of trailer heading:

$$\mathbf{n}_L = [-\sin \theta_t, \cos \theta_t]^\top, \quad (3.6)$$

then:

$$\text{ICCT} = T + R_t \mathbf{n}_L. \quad (3.7)$$

3.3.2 Intersection with Tractor Rear-Axle Trajectory

Let the infinite line through T and ICCT be:

$$\ell(t) = T + t(\text{ICCT} - T), \quad t \in \mathbb{R}. \quad (3.8)$$

Let the tractor rear-axle path be a polyline composed of segments $\overline{B_i B_{i+1}}$. For each segment, solve:

$$T + t(\text{ICCT} - T) = B_i + u(B_{i+1} - B_i), \quad (3.9)$$

where $u \in [0, 1]$ constrains the intersection to lie on the segment, while t is not constrained (infinite line). Among all valid segment intersections, choose the closest intersection point F to T :

$$F = \arg \min \|(T + t(\text{ICCT} - T)) - T\|. \quad (3.10)$$

The TF off-tracking is:

$$\text{TF} = \|T - F\|. \quad (3.11)$$

3.4 Summary

This chapter introduced the articulated kinematics and the TF off-tracking definition, which provides a more informative geometric measure for tractor–trailer path intrusion. Next, TF is extended from a scalar to a full swept-area danger zone for real-time collision warning.

Chapter 4

Swept-Area (Danger Zone) Modeling and Short-Horizon Prediction

4.1 Right-Side Body Boundary

For collision warning, a swept area of the vehicle body boundary is required. A practical representation is the right-side boundary defined by two points:

- H_r : right-side boundary near the hitch region,
- T_r : right-side boundary near the trailer rear axle.

Let the right normal of trailer heading be:

$$\mathbf{n}_R = [\sin \theta_t, -\cos \theta_t]^\top. \quad (4.1)$$

Then:

$$H_r = H + \frac{w}{2}\mathbf{n}_R, \quad T_r = T + \frac{w}{2}\mathbf{n}_R. \quad (4.2)$$

4.2 Time-Discretized Swept Area

A short prediction horizon of 1 s is used with step Δt and N steps. At each time step k , compute $(H_r(k), T_r(k))$. The swept area is approximated by a set of quadrilateral “tiles”:

$$Q_k = [H_r(k), T_r(k), T_r(k+1), H_r(k+1)], \quad k = 0, \dots, N-1. \quad (4.3)$$

The union $\bigcup_k Q_k$ forms the predicted danger zone.

4.3 Receding-Horizon Strategy in the Tractor Body Frame

To avoid global coordinate drift and simplify fusion with right-side radar, prediction is performed in the tractor rear-axle body frame (B-frame):

- origin at current B ,
- x-axis aligned with current tractor heading,
- y-axis left-positive (right side corresponds to $y < 0$).

At each control cycle (e.g., 50 Hz), the state is re-centered so that $B = (0, 0)$ and $\theta = 0$ at the start of the prediction window.

4.4 Prediction Model Used On-Board

In embedded deployment, the prediction uses current measurements (v, α, ϕ) and assumes them constant over the 1-second horizon. The predicted tractor rear-axle motion is:

$$v_B = v \cos \alpha, \tag{4.4}$$

$$x_B(k+1) = x_B(k) + v_B \cos \theta(k) \Delta t, \quad y_B(k+1) = y_B(k) + v_B \sin \theta(k) \Delta t, \tag{4.5}$$

$$\theta(k+1) = \theta(k) + \frac{v \sin \alpha}{l} \Delta t. \tag{4.6}$$

Then compute $H(k)$, $T(k)$, and boundary points $H_r(k)$ and $T_r(k)$ as in Chapter 3.

4.5 Algorithm Summary

Algorithm 1 Short-Horizon Swept Area Prediction (B-frame)

Require: Current measurements v, α, ϕ , vehicle parameters l, l_h, L, w , step Δt , horizon steps N

```
1: Initialize  $B = (0, 0)$ ,  $\theta = 0$ 
2: Compute  $H_r(0), T_r(0)$  from  $(B, \theta, \phi)$ 
3: for  $k = 0$  to  $N - 1$  do
4:    $v_B \leftarrow v \cos \alpha$ 
5:    $B \leftarrow B + v_B [\cos \theta, \sin \theta] \Delta t$ 
6:    $\theta \leftarrow \theta + \frac{v \sin \alpha}{l} \Delta t$ 
7:   Compute  $H, T$  and then  $H_r(k + 1), T_r(k + 1)$ 
8:   Define tile  $Q_k = [H_r(k), T_r(k), T_r(k + 1), H_r(k + 1)]$ 
9: end for
10: return  $\{Q_k\}_{k=0}^{N-1}$ 
```

4.6 Summary

This chapter established a computationally efficient representation of the danger zone as a union of quadrilateral tiles over a 1-second prediction horizon, enabling embedded real-time collision checking.

Chapter 5

Embedded System Design and Implementation on ESP32-S3

5.1 Hardware Overview

The embedded platform is ESP32-S3 (ESP32-S3-N16R8). The system integrates:

- Steering angle sensor (front-wheel angle) for α ,
- Articulation angle sensor for ϕ ,
- Speed source (wheel pulse or CAN-derived) for v ,
- Right-side mmWave radar providing detections (r, β) ,
- Warning actuators (buzzer/LED).

(Add a table of hardware modules and interfaces here.)

5.2 Radar Coordinate Transformation to B-frame

Radar detections are typically given in polar form:

$$x_R = r \cos \beta, \quad y_R = r \sin \beta. \quad (5.1)$$

Let the radar be mounted at position $\mathbf{t}_{BR} = [x_0, y_0]^\top$ in the B-frame and have yaw offset ψ_0 relative to the B-frame x-axis. Then:

$$\begin{bmatrix} x_B \\ y_B \end{bmatrix} = \begin{bmatrix} x_0 \\ y_0 \end{bmatrix} + \begin{bmatrix} \cos \psi_0 & -\sin \psi_0 \\ \sin \psi_0 & \cos \psi_0 \end{bmatrix} \begin{bmatrix} x_R \\ y_R \end{bmatrix}. \quad (5.2)$$

Thus each detection point is expressed in the same frame as the predicted swept tiles.

5.3 Collision Checking: Point-in-Tile Test

A detection is considered dangerous if it falls into any tile Q_k . Each quadrilateral tile can be split into two triangles:

$$Q_k = \triangle(p_0, p_1, p_2) \cup \triangle(p_0, p_2, p_3), \quad (5.3)$$

and tested with oriented area (2D cross-product) sign consistency. To reduce computation, each tile is pre-filtered by an axis-aligned bounding box (AABB) expanded by a safety margin.

5.4 Temporal Debouncing and Warning Logic

Radar detections can be intermittent due to multipath and occlusions. A debouncing strategy is applied:

- Trigger warning after N_{on} consecutive frames with a hit,
- Release warning after N_{off} consecutive frames with no hit.

This improves stability and reduces false alarms.

5.5 Software Architecture and Module Partitioning

The firmware is organized into modular components to decouple sensing, modeling, and decision-making:

- **config**: vehicle parameters, thresholds, pin mappings, radar extrinsics.
- **types**: shared data structures (vectors, targets, states).
- **sensors**: acquisition and calibration of v, α, ϕ .
- **radar**: radar frame parsing and coordinate transformation.
- **kinematics**: pure kinematic prediction (no hardware dependencies).
- **geometry**: point-in-triangle/quad and AABB tests.
- **sweep_predictor**: generation of tiles $\{Q_k\}$ for the next second.
- **alarm**: debounced warning actuator control.
- **filters**: low-pass filtering for sensor stabilization.

- `math_utils`: angle wrapping and numeric helpers.
- `main`: scheduler loop and dataflow integration.

This separation enables independent verification of the kinematic and geometry components against MATLAB results, and simplifies future substitution of sensor interfaces (ADC, I²C, SPI, CAN).

5.6 Real-Time Dataflow

At a fixed rate (e.g., 50 Hz), the system performs:

- (1) Read v, α, ϕ and apply filtering.
- (2) Predict the 1 s swept tiles in the B-frame.
- (3) Parse radar detections and transform to B-frame.
- (4) Perform point-in-tile tests with AABB acceleration.
- (5) Update debounced warning outputs.

5.7 Complexity and Feasibility

With N tiles (e.g., $N = 50$) and M radar targets per frame, complexity is $\mathcal{O}(MN)$. Each check uses simple arithmetic operations and is feasible on ESP32-S3 at 240 MHz.

5.8 Summary

This chapter presented the embedded implementation details, emphasizing frame consistency, efficient collision checking, and a modular firmware architecture suitable for iterative development.

Chapter 6

Conclusion and Future Work

6.1 Conclusion

This thesis developed a complete pipeline from MATLAB kinematic simulation to embedded real-time off-tracking warning for articulated trucks. A short-horizon swept-area model was implemented and integrated with mmWave radar detections in the tractor rear-axle body frame. The resulting system is computationally light and suitable for real-world deployment scenarios where global localization is unavailable.

6.2 Future Work

Future improvements include:

- Incorporating time-varying inputs (v, α, ϕ) using short-term extrapolation rather than constant-hold.
- Replacing simplified kinematics with models including tire slip and articulation dynamics.
- Adding target tracking and classification to reduce false positives.
- Formal evaluation with real vehicle experiments and safety metrics.

Bibliography

- [1] *(Add references here. Example: a vehicle dynamics textbook, off-tracking standards, radar specs, etc.)*

Appendix A

Supplementary Notes

A.1 Parameter Table (Fill In)

Table A.1: Vehicle and System Parameters (to be completed)

Parameter	Value	Notes
l	(<i>e.g.</i> , 4.0 m)	Tractor wheelbase
l_h	(<i>e.g.</i> , 1.8 m)	Rear overhang to hitch
L	(<i>e.g.</i> , 13.5 m)	Trailer length
w	(<i>e.g.</i> , 2.0 m)	Track width
Δt	(<i>e.g.</i> , 0.02 s)	Prediction step
N	(<i>e.g.</i> , 50)	Horizon steps

A.2 Figure Placeholders

Placeholder for a system block diagram.

Figure A.1: System block diagram (to be inserted).